

Cellula Technologies - LSTM Training Results

One Week Task - Multiclass Text Classification

95.33%	0.9504	0.9513	>= 0.85 (PASS)
Accuracy	Macro F1	Weighted F1	Target

Executive Summary

The submitted LSTM uses a Bidirectional LSTM over trainable word embeddings. The model was evaluated on a 300-row stratified test split. It achieved a Macro F1 of **0.9504**, which exceeds the assignment target of **0.85**.

Training used EarlyStopping and ReduceLROnPlateau. The best validation-loss weights were restored from epoch 5; training stopped at epoch 10.

Dataset and Preprocessing

- Dataset size: 3,000 rows and 9 target classes.
- Input features: query text + image description text.
- Split: 80% train (2,400), 10% validation (300), 10% test (300), stratified by class.
- Vocabulary size: 3,972 tokens; padded sequence length: 44.
- Class imbalance was handled with clipped class weights in the range 0.6 to 3.0.
- The tokenizer was fitted on the training split only.

Data Quality Findings

Finding	Observed value
Exact duplicate rows	973
Rows in repeated-query groups	1,000
Known template-like rows	1,000 (33.33%)
Unique image descriptions	12
Validation exact overlap with Train	90 / 300 (30.00%)
Test exact overlap with Train	114 / 300 (38.00%)

Interpretation: The benchmark result can be optimistic because repeated/template-like inputs can occur in both training and test partitions. The reported score is therefore best interpreted as performance on the supplied assignment benchmark rather than a leakage-free estimate on completely novel production data.

LSTM Architecture

Layer	Output Shape	Parameters
Embedding	(None, 44, 128)	508,416
SpatialDropout1D	(None, 44, 128)	0
Bidirectional LSTM	(None, 44, 256)	263,168
GlobalMaxPooling1D	(None, 256)	0
Dense (ReLU)	(None, 128)	32,896
Dropout	(None, 128)	0
Dense (Softmax)	(None, 9)	1,161
Total		805,641

Training Curves



The model converged quickly. Validation performance stabilized earlier than training performance, and EarlyStopping restored the best validation-loss checkpoint.

Training Configuration

Setting	Value
Optimizer	Adam
Initial learning rate	0.0005
Batch size	32
Maximum epochs	35
EarlyStopping patience	5
ReduceLROnPlateau factor	0.5
Loss	Sparse categorical cross-entropy

Final Test Results

95.33%	0.9504	0.9513	>= 0.85 (PASS)
Accuracy	Macro F1	Weighted F1	Target

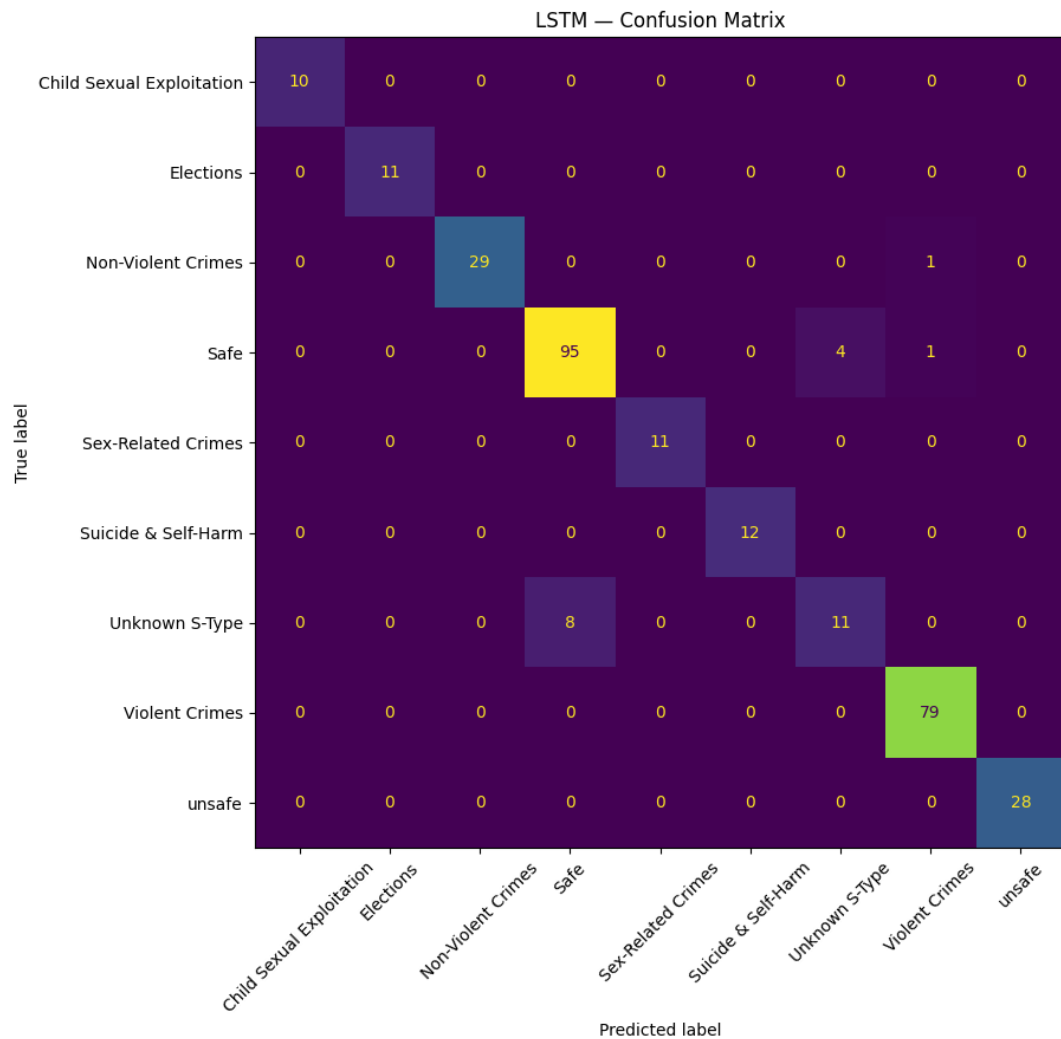
Per-Class Classification Report

Class	Precision	Recall	F1	Support
Child Sexual Exploitation	1.0000	1.0000	1.0000	10
Elections	1.0000	1.0000	1.0000	11
Non-Violent Crimes	1.0000	0.9667	0.9831	30
Safe	0.9223	0.9500	0.9360	100
Sex-Related Crimes	1.0000	1.0000	1.0000	11
Suicide & Self-Harm	1.0000	1.0000	1.0000	12
Unknown S-Type	0.7333	0.5789	0.6471	19
Violent Crimes	0.9753	1.0000	0.9875	79
unsafe	1.0000	1.0000	1.0000	28

Result Interpretation

The lowest per-class F1 is for **Unknown S-Type** at **0.6471**. This class remains the main source of classification difficulty. The LSTM improves this difficult class compared with the RNN while also improving the overall benchmark score.

Confusion Matrix



Conclusion

The LSTM model **passes** the assignment requirement with Macro F1 = **0.9504** versus the required **0.85**. For production use, a manually verified and deduplicated dataset, group-aware splitting, and a genuinely unseen external holdout set are recommended.